Name → Harleen kaur
UID → 23BCS10342
Section → KRG 3A

Experiment → 5

Test Cases

2 3
a a a a a   O/P → yes
a c a c a c a   no
a a b a a   no
c c a c a c c
c a a a c

Ques → Watto, the owner of a spare parts store, has recently got an order for the mechanism.
Given string s, determine if the memory of the mechanism contains string t, that consists of the same number of characters as s and differs from s in exactly one position.

⟹ Approach

1) Store all memory strings in a set.
2) For each Query string :
   - Try changing each position
   - Replace it with other two possible characters.
   - Check if modified string exists in set.
3) Because :
   - only 3 characters → at each index only 2 replacements possible.
   - If length = L → total checks = 2L
   - Total operations = O(total length) = O(L).

⟹ Code

```
#include <bits/stdc++.h>
using namespace std;

int main {
```

```
int n, m;
cin >> n >> m;

unordered_set<string> st
for (int i=0; i<n; i++) {
  string s;
  cin >> s;
  st.insert(s);
}
while (m--) {
  string s;
  cin >> s;
  bool found = false;
  for (int i=0; i<s.length() && !found; i++) {
    char original = s[i];
    for (char ch='a'; ch<='c'; ch++) {
      if (ch == original) continue;
      s[i] = ch;

      if (st.count(s)) {
        found = true;
        break;
      }
    }
    s[i] = original;
  } ;
  if (found) {
    cout << "Yes" << endl;
  }
  else {
  cout << "No" << endl;
}
}
```